

The distributive law



- 1 A student incorrectly used the distributive property and wrote $6(7x + 2) = 42x + 2$.

Explain how to correct the error.

- 2 Expand the following expressions:

a $5(s + 7)$	b $9(5 + w)$	c $3(r - 6)$	d $(y + 8) \times 7$
e $-8(t + 10)$	f $-8(c - 5)$	g $-(y + 2)$	h $4(4s + 5)$
i $-10(8 + 3r)$	j $-8(-2t - 3)$	k $p(q + 4)$	l $9(u - v)$
m $-(2m - 3n)$	n $-r(11s + 13t)$	o $4(2r + 3s + 5t)$	p $6(4m + 3n + 5p)$

- 3 Expand the following expressions:

a $6m(2n + 3)$	b $9(v^2 + 6)$	c $r(r + 7)$	d $6m(4m + 7)$
e $-6a(2a + 9)$	f $4wy(y + w)$	g $4p(4q - 3 + 6r)$	h $2f(f - 7g - 6h)$
i $-10(y^2 + 7)$	j $-(9 - s^3)$	k $r(s^2 + 8)$	l $-10(3 + 8u^3)$
m $4s^5(4t^4 - 9)$	n $6p^3(2q^5 - 9)$	o $9f(f - 4g - 6h)$	p $3uv(v + u)$

- 4 Expand and simplify the following expressions:

a $4(x + 8) - 2$	b $7 + 5(3x + 4)$
c $5 + 3(x + 4)$	d $8(3x + 4) - 6x$
e $34 + 9(4x - 5)$	f $5x - 8(2x - 3)$
g $9x - 5(2x + 3)$	h $-8(5x - 7) - 9$
i $4y + 5 + 6(y - 9)$	j $4(5(x - 3) + 2)$

- 5 Expand and simplify the following expressions:

a $x(x + 4) + 8(x + 9)$	b $8y^2 - 4y + 7y(y - 5)$
c $3(6x - 5y) - 2(9y + 4x)$	d $7(y + 9) + 5(y + 6)$
e $10(x - 4) + 3(x + 8)$	f $5(y + 7) - (y - 9)$
g $-8(x + 4) - 3(x - 2)$	h $11(a + 4) - 6(4 - a)$
i $4u(6u + 9v) - 2u(8v + 5u)$	j $9w(8w + 6z) - 5w(4z + 2w)$

Applications

6 A color printer company offers a range of color panels each with a resolution of

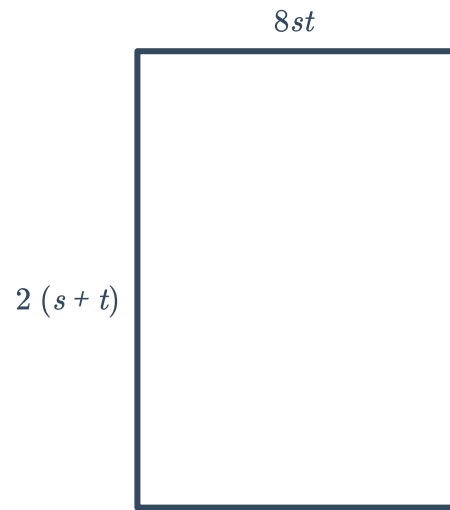
$(m + 8)$ kW, where m is the panel model number. Write an expression in terms of m for the total amount of power produced by 11 solar panels.



- 7 The representatives of 8 countries are attending a global summit, and each country is able to send $(2r - 3)$ representatives. Write an expression for the total number of representatives attending the summit.
- 8 A forklift operator has enough time to make 5 trips between the two ends of a warehouse. A forklift can carry $(2 + 3n^4)$ boxes at a time. Write an expression for the total number of boxes that can be carried across the warehouse.

- 9 Consider the rectangle shown:

Find an expression for the area of the rectangle in expanded form.



- 10 A rectangle is 8 m longer than it is width. Its width is represented by the pronumeral u . Form an expression for the area of the rectangle in expanded form.